Math 241, F12
Exam 1

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	25	
2	25	
3	30	
4	20	
Total	100	

Good Luck !

Problem 1. (25 points) Given two vectors $\mathbf{a} =< 3,\ \sqrt{3},\ 2\sqrt{6} >$ and $\mathbf{b} =< 0,\ -\sqrt{3},\ \sqrt{6} >$.

9 (a) Find $|\mathbf{a}|$, $|\mathbf{b}|$, and $\mathbf{a}\cdot\mathbf{b}$.

$$|a| = \sqrt{3^2+(\sqrt{3})^2+(2\sqrt{6})^2} = \sqrt{9+3+24} = \sqrt{36} = 6$$

$$|b| = \sqrt{0^2+(-\sqrt{3})^2+(\sqrt{6})^2} = \sqrt{0+3+6} = \sqrt{9} = 3$$

$$a\cdot b = 3\cdot 0 + (\sqrt{3})\cdot(-\sqrt{3}) + 2\sqrt{6}\cdot\sqrt{6} = 0+(-3)+12 = 9$$

4 (b) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{b}$.

$$u = \frac{b}{|b|} = \frac{<0, -\sqrt{3}, \sqrt{6}>}{3} = <0, -\frac{\sqrt{3}}{3}, \frac{\sqrt{6}}{3}>$$

6 (c) Find the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

$$\cos\theta = \frac{a\cdot b}{|a|\,|b|}$$

$$= \frac{9}{3\cdot 6} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

6 (d) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, $\text{proj}_{\mathbf{a}}\mathbf{b}$.

$$\text{Proj}_a b = \left(\frac{a\cdot b}{|a|^2}\right) a$$

$$= \left(\frac{9}{6^2}\right)\cdot <3, \sqrt{3}, 2\sqrt{6}>$$

$$= \frac{1}{4} <3, \sqrt{3}, 2\sqrt{6}>$$

$$= <\frac{3}{4}, \frac{\sqrt{3}}{4}, \frac{\sqrt{6}}{2}>$$

Problem 2. (25 points) Given three points $P(1,0,1)$, $Q(-1,3,-2)$, and $R(3,0,5)$.

8 (a) Find parametric equations and symmetric equations for the line through the points P and Q.

$$V = \overrightarrow{PQ} = \langle -1,3,-2\rangle - \langle 1,0,1\rangle = \langle -2,3,-3\rangle$$

Parametric equations:

$$x = 1-2t, \quad y = 3t, \quad z = 1-3t$$

symmetric equations:

$$\frac{x-1}{-2} = \frac{y}{3} = \frac{z-1}{-3}$$

12 (b) Find an equation for the plane that contains the triangle $\triangle PQR$.

$$\overrightarrow{PQ} = \langle -2,3,-3\rangle$$

$$\overrightarrow{PR} = \langle 2,0,4\rangle$$

$$n = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} i & j & k \\ -2 & 3 & -3 \\ 2 & 0 & 4 \end{vmatrix} = 12i + 2j - 6k$$

An equation is given by:

$$12(x-1) + 2(y-0) - 6(z-1) = 0$$

$$12x + 2y - 6z = 6$$

$$\underline{6x + y - 3z = 3}$$

5 (c) Find the area of the triangle $\triangle PQR$.

$$\begin{aligned} \text{Area}(\triangle PQR) &= \frac{1}{2}\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| \\ &= \frac{1}{2}\left|12i + 2j - 6k\right| \\ &= \frac{1}{2}\sqrt{12^2 + 2^2 + (-6)^2} \\ &= \frac{1}{2}\sqrt{184} \\ &= \sqrt{46} \end{aligned}$$

Problem 3. (30 points)

12 (a) Let $\mathbf{r}(t) = 2t\mathbf{i} + t^2\mathbf{j} - \frac{1}{3}t^3\mathbf{k}$, find the arc length of the curve defined by $\mathbf{r}(t)$ with $0 \le t \le 1$.

$$r'(t) = 2i + 2tj - t^2k$$

$$\Rightarrow |r'(t)| = \sqrt{2^2+(2t)^2+(-t^2)^2} = \sqrt{4+4t^2+t^4}$$

$$= \sqrt{(t^2+2)^2} = t^2+2$$

Thus

$$L = \int_0^1 |r'(t)|\,dt$$

$$= \int_0^1 t^2+2\,dt$$

$$= \left(\frac{t^3}{3}+2t\right)\Big|_{t=0}^{t=1}$$

$$= \left(\frac{1}{3}+2\right) - 0 = \frac{7}{3}$$

18 (b) Let $\mathbf{r}(t) = < t,\ e^t \cos t, e^t \sin t >$. Find the tangent unit vector $\mathbf{T}$ and the curvature κ of the curve at the point with $t = 0$.

$$r'(t) = \langle 1, e^t\cos t - e^t\sin t,\ e^t\sin t + e^t\cos t\rangle = \langle 1, e^t(\cos t - \sin t), e^t(\cos t + \sin t)\rangle$$

$$\text{So } r'(0) = \langle 1, e^0(\cos 0 - \sin 0), e^0(\cos 0 - \sin 0)\rangle$$

$$= \langle 1, 1, 1\rangle$$

$$|r'(0)| = \sqrt{1^2+1^2+1^2} = \sqrt{3}$$

$$\Rightarrow T(0) = \frac{r'(0)}{|r'(0)|} = \frac{1}{\sqrt{3}}\langle 1, 1, 1\rangle$$

$$= \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right\rangle$$

$$r''(t) = \langle 0, e^t(\cos t - \sin t) + e^t(-\sin t - \cos t), e^t(\cos t + \sin t) + e^t(-\sin t + \cos t)\rangle$$

$$= \langle 0, -2e^t\sin t, 2e^t\cos t\rangle$$

$$\text{So } r''(0) = \langle 0, -2e^0\sin 0, 2e^0\cos 0\rangle = \langle 0, 0, 2\rangle$$

$$r'(0) \times r''(0) = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = 2i - 2j = \langle 2, -2, 0\rangle$$

$$|r'(0) \times r''(0)| = \sqrt{2^2+(-2)^2+0^2} = \sqrt{8} = 2\sqrt{2}$$

$$\Rightarrow k(0) = \frac{|r'(0)\times r''(0)|}{|r'(0)|^3} = \frac{2\sqrt{2}}{(\sqrt{3})^3} = \frac{2\sqrt{2}}{3\sqrt{3}} = \frac{2\sqrt{6}}{9}$$

Problem 4. (20 points)

12 (a) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has the acceleration $\mathbf{a}(t) = 2t\mathbf{i} + e^t\mathbf{j} + e^{-t}k$ and initial conditions $\mathbf{v}(0) = \mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{r}(0) = \mathbf{i}$.

$$V(t) = \int a(t)dt + C_1 = \int \langle 2t, e^t, e^{-t}\rangle dt + C_1$$
$$= \langle t^2, e^t, -e^{-t}\rangle + C_1$$
$$V(0) = \langle 1, 1, -1\rangle \Rightarrow \langle 0, 1, -1\rangle + C_1 = \langle 1, 1, -1\rangle$$
$$\Rightarrow C_1 = \langle 1, 1, -1\rangle - \langle 0, 1, -1\rangle = \langle 1, 0, 0\rangle$$
Thus $V(t) = \langle t^2, e^t, -e^{-t}\rangle + \langle 1, 0, 0\rangle$
$$= \langle t^2+1, e^t, -e^{-t}\rangle$$
$$r(t) = \int v(t)dt + C_2 = \int \langle t^2+1, e^t, -e^{-t}\rangle dt + C_2$$
$$= \langle \frac{t^3}{3} + t, e^t, e^{-t}\rangle + C_2$$
$$r(0) = \langle 1, 0, 0\rangle \Rightarrow \langle 0, 1, 1\rangle + C_2 = \langle 1, 0, 0\rangle$$
$$\Rightarrow C_2 = \langle 1, 0, 0\rangle - \langle 0, 1, 1\rangle = \langle 1, -1, -1\rangle$$
Thus $r(t) = \langle \frac{t^3}{3} + t, e^t, e^{-t}\rangle + \langle 1, -1, -1\rangle$
$$= \langle \frac{t^3}{3} + t + 1, e^t - 1, e^{-t} - 1\rangle$$

8 (b) Find the angle between the two planes $x + y + z = 1$ and $x - 2y - 3z = 3$.

$$n_1 = \langle 1, 1, 1\rangle$$
$$n_2 = \langle 1, -2, -3\rangle$$
$$\cos\theta = \frac{|n_1 \cdot n_2|}{|n_1|\,|n_2|} = \frac{|1\cdot1 + 1\cdot(-2) + 1\cdot(-3)|}{\sqrt{1^2+1^2+1^2}\cdot\sqrt{1^2+(-2)^2+(-3)^2}}$$
$$= \frac{|(-4)|}{\sqrt{3}\cdot\sqrt{14}} = \frac{4}{\sqrt{42}}$$
$$\Rightarrow \theta = \cos^{-1}\frac{4}{\sqrt{42}}$$